Centre Number	Candidate Number	Name

UNIVERSITY OF CAMBRIDGE INTERNATIONAL EXAMINATIONS
International General Certificate of Secondary Education

CHEMISTRY **0620/02**

Paper 2

May/June 2004

1 hour 15 minutes

Candidates answer on the Question Paper.
No Additional Materials required.

READ THESE INSTRUCTIONS FIRST

Write your Centre number, candidate number and name on all the work you hand in.
Write in dark blue or black pen in the spaces provided on the Question Paper.
You may use a pencil for any diagrams, graphs or rough working.
Do not use staples, paper clips, highlighters, glue or correction fluid.

Answer **all** questions.
The number of marks is given in brackets [ ] at the end of each question or part question.
A copy of the Periodic Table is printed on page 16.

If you have been given a label, look at the details. If any details are incorrect or missing, please fill in your correct details in the space given at the top of this page.

Stick your personal label here, if provided.

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2	
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Total	

This document consists of **16** printed pages.

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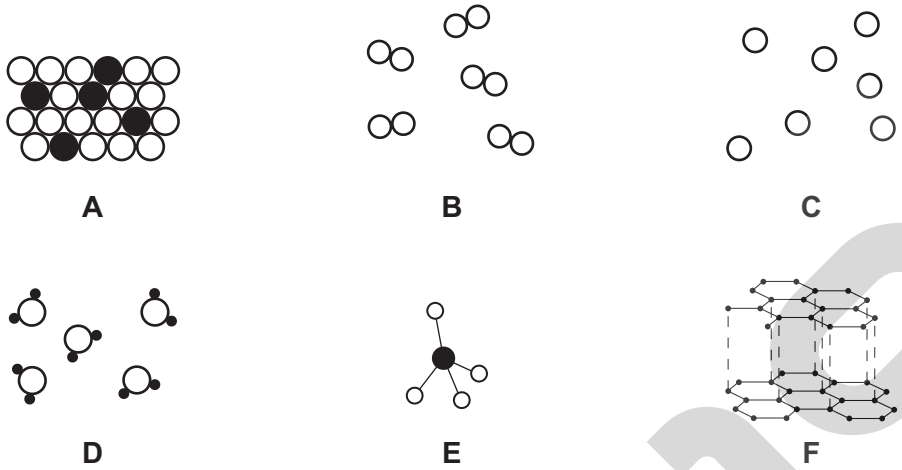
1 The diagram shows models of various structures,



(a) Which **three** of the structures **A** to **F** represent elements? Give a reason for your answer.

structures .....

reason ..... [2]

(b) Which one of the structures **A** to **F** represents a gas containing single atoms?

..... [1]

(c) **(i)** Which one of the structures **A** to **F** represents a gas containing diatomic molecules?

.....

(ii) State the name of a gas which has diatomic molecules.

..... [2]

(d) **(i)** Which one of the structures **A** to **F** represents graphite?

.....

(ii) State **one** use of graphite.

..... [2]

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(e) Structure **D** represents a compound.

(i) State what is meant by the term *compound*.

.....

.....

(ii) Which one of the following substances is structure **E** most likely to represent?

Put a ring around the correct answer.

ammonia **hydrogen chloride** **methane** **water** [2]

(f) Hydrogen chloride is a compound.

(i) Draw a diagram to show how the electrons are arranged in a molecule of hydrogen chloride.

Show only the outer electrons.

show hydrogen electrons as •
show chlorine electrons as x

..... [2]

(ii) State the name of the type of bonding present in hydrogen chloride.

..... [1]

(iii) Hydrogen chloride dissolves in water to form an acidic solution (hydrochloric acid). Describe how you would use litmus paper to show that this solution is acidic.

.....

..... [2]

(iv) Which one of the following values is most likely to represent the pH of a dilute solution of hydrochloric acid?

Put a ring around the correct answer.

pH 2 **pH7** **pH10** **pH14** [1]

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(v) Complete the following equation for the reaction of hydrochloric acid with magnesium.

$Mg(s) + \ldots\ldots\ldots\ldots HCl(aq) \rightarrow MgCl_2(aq) + H_2(g)$ [1]

(vi) Name the salt formed in this reaction.

..... [1]

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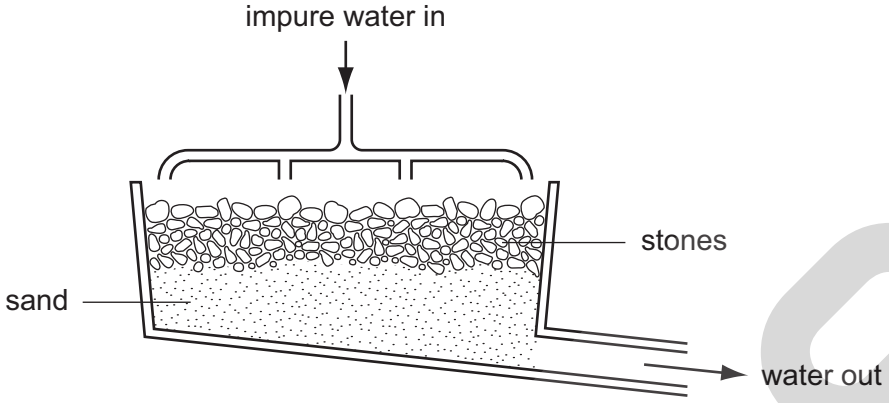
2 Two of the stages in water purification are filtration and chlorination.
The diagram below shows a filter tank.



(a) Explain how this filter helps purify the water.

.....

.....

..... [2]

(b) **(i)** Why is chlorine added during water purification?

.....

(ii) After chlorination, the water is acidic. A small amount of slaked lime is added to the acidic water. Explain why slaked lime is added.

.....

(iii) What is the chemical name for slaked lime?

.....

(iv) State **one** other use of slaked lime.

..... [4]

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(c) **(i)** State the boiling point of pure water.

..... [2]

(ii) Describe a chemical test for water.

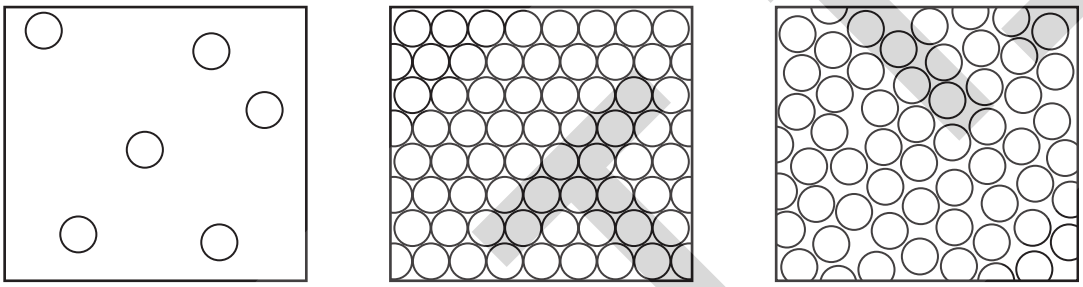
test ..... [1]

result ..... [1]

(iii) State **one** use of water in the home.

..... [1]

(d) The diagram shows the arrangement of particles in the three different states of water.



A **B** **C**

Which of these diagrams, **A**, **B** or **C**, shows water in a solid state?

..... [1]

(e) Steam reacts with ethene in the presence of a catalyst. Complete the word equation for this reaction.

ethene + steam → ..... [1]

(f) Potassium reacts violently with water. Complete the word equation for this reaction.

potassium + water → ..... + ...

..... [2]

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3 When lumps of calcium carbonate react with hydrochloric acid, carbon dioxide gas is released.

$$CaCO_3(s) + 2HCl(aq) \rightarrow CaCl_2(aq) + CO_2(g) + H_2O(l)$$

(a) Describe a practical method for investigating this reaction, which would enable you to calculate the rate of reaction.

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

..... [4]

(b) What effect will the following have on the rate of the reaction?

(i) increasing the temperature

.....

(ii) adding water to the acid

.....

(iii) using powdered calcium carbonate instead of lumps

..... [3]

(c) Describe a test for calcium ions.

result .....

test .....

..... [3]

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(d) Calcium can be obtained by the electrolysis of molten calcium chloride.

(i) Suggest why calcium must be extracted by electrolysis rather than by reduction with carbon.

..... [1]

(ii) Draw the electronic structure of an atom of calcium.

[2]

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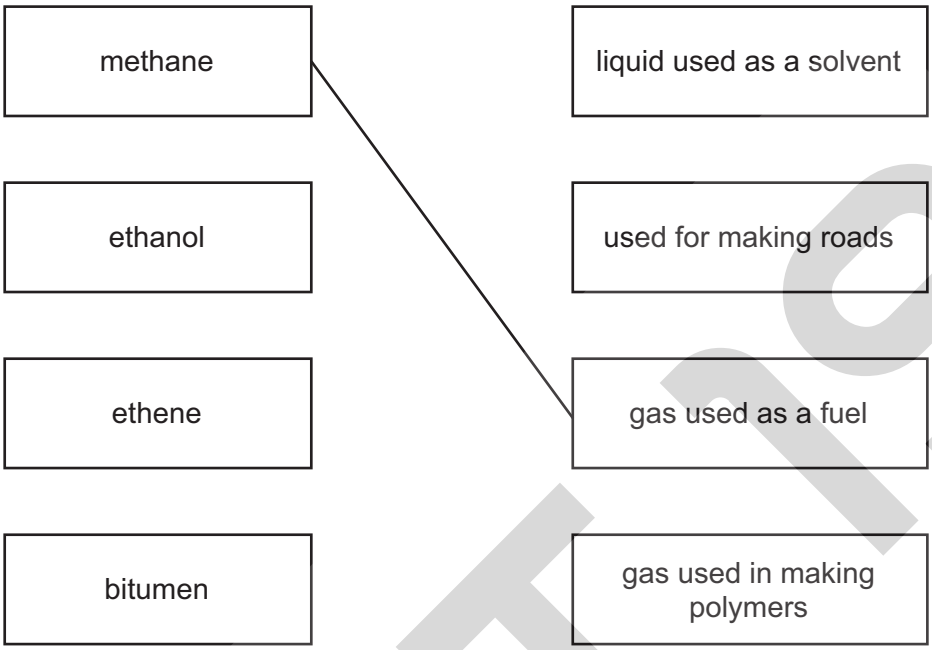
4 Organic substances have many uses.

(a) Match the substances in the boxes on the left with the descriptions in the boxes on the right. The first one has been done for you.



[3]

(b) Which one of the following would be least likely to be obtained from the fractional distillation of petroleum? Put a ring around the correct answer.

bitumen **ethane** **ethanol** **methane** [1]

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(c) Some reactions of organic compounds are shown below.

A $n\ CH_2{=}CH_2 \longrightarrow \left(CH_2 - CH_2 \right)_n$

B $C_3H_8 + 5O_2 \longrightarrow 3CO_2 + 4H_2O$

C $C_6H_{12}O_6$ glucose $\longrightarrow 2CO_2 + 2C_2H_5OH$

D $C_8H_{18} \longrightarrow C_6H_{14} + C_2H_4$

(i) Which **one** of the reactions, **A**, **B**, **C** or **D**, shows fermentation?

.....

(ii) Which **one** of the reactions, **A**, **B**, **C** or **D**, shows polymerization?

.....

(iii) Which **one** of the reactions, **A**, **B**, **C** or **D**, shows combustion?

.....

(iv) Which **one** of the reactions, **A**, **B**, **C** or **D**, shows cracking?

..... [4]

(d) The hydrocarbon C_8H_{18} is an alkane.

(i) What is meant by the term *hydrocarbon*?

.....

(ii) Explain why this hydrocarbon is an alkane.

..... [2]

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5 Look at the list of five elements below.

argon
bromine
chlorine
iodine
potassium

(a) Put these five elements in order of increasing proton number.

..... [1]

(b) Put these five elements in order of increasing relative atomic mass.

..... [1]

(c) The orders of proton number and relative atomic mass for these five elements are different. Which **one** of the following is the most likely explanation for this?

Tick **one** box.

The proton number of a particular element may vary. ☐

The presence of neutrons. ☐

The atoms easily gain or lose electrons. ☐

The number of protons must always equal the number of neutrons. ☐

[1]

(d) Which of the five elements in the list are in the same group of the Periodic Table?

..... [1]

(e) **(i)** From the list, choose **one** element which has one electron in its outer shell.

..... [1]

(ii) From the list, choose **one** element which has a full outer shell of electrons.

..... [1]

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(f) Which **two** of the following statements about argon are correct?

Tick **two** boxes.

Argon is a noble gas. ☐

Argon reacts readily with potassium. ☐

Argon is used to fill weather balloons. ☐

Argon is used in light bulbs. ☐

[2]

(g) Potassium chloride can be made by reacting potassium with chlorine. The bonding in potassium chloride is ionic.

What does this information tell you about

(i) the boiling point of potassium chloride,

..... [1]

(ii) the electrical conductivity of molten potassium chloride?

..... [1]

(h) Describe the change in the electronic structure of potassium and chlorine atoms when they combine to make potassium chloride.

change in potassium atom

.....

change in chlorine atom

..... [2]

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6 Iron is extracted from its ore in a blast furnace using carbon (coke) as a reducing agent and as a source of heat.

(a) The coke burns in hot air. The equation for this reaction is

$$2C(s) + O_2(g) \rightarrow 2CO(g)$$

State the name of the gas produced in this reaction.

..... [1]

(b) Near the top of the blast furnace, the iron(III) oxide in the iron ore gets reduced to iron.

$$Fe_2O_3(s) + 3CO(g) \rightarrow 2Fe(l) + 3CO_2(g)$$

Use the equation to explain why the change of iron(III) oxide to iron is a reduction reaction.

..... [1]

(c) In the hottest regions of the furnace, iron(III) oxide is reduced by carbon. Complete the equation for this reaction.

$Fe_2O_3(s)$ + ...C(s) → .....Fe(l) + 3CO(g) [2]

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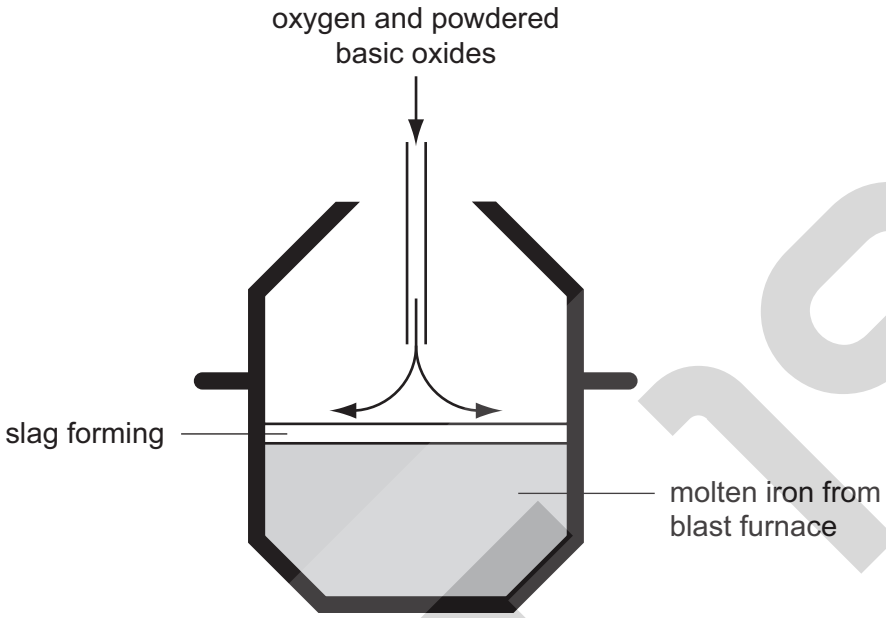
(d) The iron from the blast furnace contains up to 10% by mass of impurities. The main impurities are carbon, silicon and phosphorus. The diagram below shows one method of making steel from iron.



A mixture of oxygen and basic oxides is blown onto the surface of the molten iron.

(i) What is the purpose of blowing oxygen onto the molten iron?

..... [1]

(ii) A large amount of energy is released in the process of steelmaking.
What name is given to chemical reactions which release energy?

..... [1]

(iii) The basic oxides react with the impurities in the iron and form a slag. What information in the diagram suggests that the slag is less dense than the molten iron?

..... [1]

(iv) Which one of the following is a basic oxide?
Put a ring around the correct answer.

calcium oxide **carbon dioxide** **sulphur dioxide** **water** [1]

(v) Why is steel rather than iron used for constructing buildings and bridges?

..... [1]

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(e) Special steels contain added elements such as vanadium, chromium, cobalt or nickel. These are all transition metals.

State three properties of transition metals which are **not** shown by non-transition metals.

1. .....

2. .....

3. ..... [3]

(f) What is the name given to metals which are mixtures of more than one metal?

..... [1]

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DATA SHEET
The Periodic Table of the Elements

Group

I	II											III	IV	V	VI	VII	0
							1 **H** Hydrogen 1										4 **He** Helium 2
7 **Li** Lithium 3	9 **Be** Beryllium 4											11 **B** Boron 5	12 **C** Carbon 6	14 **N** Nitrogen 7	16 **O** Oxygen 8	19 **F** Fluorine 9	20 **Ne** Neon 10
23 **Na** Sodium 11	24 **Mg** Magnesium 12											27 **Al** Aluminium 13	28 **Si** Silicon 14	31 **P** Phosphorus 15	32 **S** Sulphur 16	35.5 **Cl** Chlorine 17	40 **Ar** Argon 18
39 **K** Potassium 19	40 **Ca** Calcium 20	45 **Sc** Scandium 21	48 **Ti** Titanium 22	51 **V** Vanadium 23	52 **Cr** Chromium 24	55 **Mn** Manganese 25	56 **Fe** Iron 26	59 **Co** Cobalt 27	59 **Ni** Nickel 28	64 **Cu** Copper 29	65 **Zn** Zinc 30	70 **Ga** Gallium 31	73 **Ge** Germanium 32	75 **As** Arsenic 33	79 **Se** Selenium 34	80 **Br** Bromine 35	84 **Kr** Krypton 36
85 **Rb** Rubidium 37	88 **Sr** Strontium 38	89 **Y** Yttrium 39	91 **Zr** Zirconium 40	93 **Nb** Niobium 41	96 **Mo** Molybdenum 42	**Tc** Technetium 43	101 **Ru** Ruthenium 44	103 **Rh** Rhodium 45	106 **Pd** Palladium 46	108 **Ag** Silver 47	112 **Cd** Cadmium 48	115 **In** Indium 49	119 **Sn** Tin 50	122 **Sb** Antimony 51	128 **Te** Tellurium 52	127 **I** Iodine 53	131 **Xe** Xenon 54
133 **Cs** Caesium 55	137 **Ba** Barium 56	139 **La** Lanthanum 57 *	178 **Hf** Hafnium 72	181 **Ta** Tantalum 73	184 **W** Tungsten 74	186 **Re** Rhenium 75	190 **Os** Osmium 76	192 **Ir** Iridium 77	195 **Pt** Platinum 78	197 **Au** Gold 79	201 **Hg** Mercury 80	204 **Tl** Thallium 81	207 **Pb** Lead 82	209 **Bi** Bismuth 83	**Po** Polonium 84	**At** Astatine 85	**Rn** Radon 86
Fr Francium 87	226 **Ra** Radium 88	227 **Ac** Actinium 89															

140 **Ce** Cerium 58	141 **Pr** Praseodymium 59	144 **Nd** Neodymium 60	**Pm** Promethium 61	150 **Sm** Samarium 62	152 **Eu** Europium 63	157 **Gd** Gadolinium 64	159 **Tb** Terbium 65	162 **Dy** Dysprosium 66	165 **Ho** Holmium 67	167 **Er** Erbium 68	169 **Tm** Thulium 69	173 **Yb** Ytterbium 70	175 **Lu** Lutetium 71
232 **Th** Thorium 90	**Pa** Protactinium 91	238 **U** Uranium 92	**Np** Neptunium 93	**Pu** Plutonium 94	**Am** Americium 95	**Cm** Curium 96	**Bk** Berkelium 97	**Cf** Californium 98	**Es** Einsteinium 99	**Fm** Fermium 100	**Md** Mendelevium 101	**No** Nobelium 102	**Lr** Lawrencium 103

*58-71 Lanthanoid series
90-103 Actinoid series

Key

a
X
b

a = relative atomic mass
X = atomic symbol
b = proton (atomic) number

The volume of one mole of any gas is 24 dm^3 at room temperature and pressure (r.t.p.).

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